

An Equivalent-Key Chosen-Plaintext Break of a 2025 Chaos-Based Image Cipher

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Abstract—Chaos-based image ciphers are published at a high rate and validated by statistical scores that do not measure resistance to a chosen-plaintext attack. We break a 2025 chaos-based image cipher (Jain *et al.*, *Optik*; Lorenz/logistic maps, “LLEO”) that reports near-ideal entropy (7.9914) and a 99.6% NPCR. From 4 chosen plaintexts we recover a key-free decryptor and decrypt any 512×512 ciphertext pixel-exactly in 0.037 s, without the key. The cause is structural: LLEO’s keystream and both permutations depend only on the secret key, never on the image, so the whole cipher is one fixed map. Its reported differential metric is then an illusion: with no plaintext diffusion the *true* one-pixel NPCR is 0.000381 % ($100/MN$), not 99.6%. A reusable structural audit and a graded security metric are developed in an extended version. Good statistics are not security.

Keywords—cryptanalysis, image encryption, chaotic cipher, chosen-plaintext attack, equivalent key, NPCR

I. INTRODUCTION

Chaos-based image ciphers appear in the literature at a high rate, and a recurring lesson from the cryptanalysis side is that a large fraction of them succumb to the same structural attack. Most follow a permutation–diffusion template: a confusion stage that rearranges pixels and a diffusion stage that masks pixel values with a chaotic keystream. When that keystream is a function of the secret key alone and not of the image, the entire cipher collapses to a *fixed* permutation–substitution map, and an *equivalent key* (a stand-in that decrypts every ciphertext under the same secret key, without being the key itself) can be recovered from a handful of chosen plaintexts (input images the attacker picks and has encrypted) [1]–[4]. The defence is equally well known: the keystream must depend on the plaintext, and the cipher must actually diffuse [5].

A second, persistent problem is methodological. Authors validate schemes with statistical scores: Shannon entropy (a measure of randomness), adjacent-pixel correlation (how alike neighbouring pixels are), and the differential metrics NPCR and UACI (how much the output changes when one input pixel changes), and treat near-ideal values as evidence of security. None of these scores measures resistance to a chosen-plaintext attack, and, as we show, all of them can be near-ideal for a cipher that is broken in seconds; these statistical tests are known to be necessary but not sufficient [6].

Why this matters operationally. Image ciphers sit in the middle of real decisions: protected medical imaging, sealed evidence, access-controlled documents. Chaos-based image ciphers in particular are proposed as the confidentiality

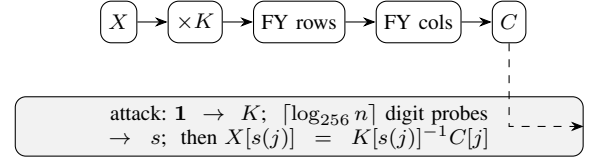


Fig. 1. LLEO encryption (top): key-only multiplicative substitution $\times K$, then two Fisher–Yates block shuffles. The equivalent-key attack (bottom) recovers the multiplier K and source map s from a few chosen plaintexts and inverts the cipher without the key.

layer for exactly these sensitive streams, most prominently telemedicine and DICOM medical imaging [7]. A break is therefore a standing capability to decrypt every future ciphertext of the probed size, so a security argument should rest on an attack model rather than a statistics table. The quantity that matters is not how noise-like a single ciphertext looks but whether *any* value in the cipher depends on the image, the one binary property that separates “broken in 4 queries” from “resists.”

A. Real-world stakes

Chaos-based image encryption is a large, active design literature, published at high volume and aimed at the most sensitive, most resource-constrained image streams: medical and DICOM imaging, the Internet of Medical Things, IoT-edge and surveillance pipelines, and satellite links, with real-time FPGA and hardware realisations [8]. No publicly documented product is yet known to rely on such a cipher, these designs are not standardised by ISO/IEEE/NIST [9], and mainstream systems still use vetted primitives such as AES; they are proposed *as lightweight replacements* for exactly those settings, on a motivation whose security benefit has itself been questioned [6], which makes their security a practical question rather than an academic one. A structural, few-query break of a single scheme is therefore a concrete warning about a technology being advanced toward that field, and a reminder that security here should be argued from an attack model, not a statistics table.

This paper applies both lessons to a 2025 scheme [10] (which we abbreviate LLEO) built from the Lorenz [11] and logistic maps with Fisher–Yates block shuffling; Fig. 1 sketches both its three-step encryption and our equivalent-key attack. Our contributions are:

- We identify that LLEO is entirely key-driven and therefore a fixed monomial map over $\mathbb{Z}_{256}$: each output pixel is one input pixel times a fixed multiplier, with all arithmetic taken modulo 256 (wrapping around like a

clock) (Section III–IV).

- We give an equivalent-key chosen-plaintext attack that recovers the full map in $1 + \lceil \log_{256} MN \rceil$ chosen plaintexts (4 for a 512×512 image) and decrypts any image pixel-exactly without the key (Section IV).
- We reproduce LLEO, confirm it matches the paper’s reported statistics, and then show those statistics coexist with *zero* plaintext diffusion: the true one-pixel NPCR is 0.000381 %, not 99.6 % (Section V).
- Most importantly, we distil the attack into a reusable, black-box *structural audit* that decides plaintext-independence for any cipher, with a held-out certificate we prove sound; the audit, its application to real published 2023–2025 schemes, and a graded plaintext-keying-capacity metric are developed in an extended version.

Fig. 2 previews the outcome: four standard images, their LLEO ciphertexts, and our key-free recoveries, each bit-identical to the original and obtained in 4 chosen plaintexts.

II. RELATED WORK

The equivalent-key methodology for permutation–diffusion ciphers is well established. Arroyo *et al.* [3] broke a “total shuffling” scheme by recovering its key-only keystream; Li *et al.* [2] recovered an equivalent key for an entropy-based cipher; and the survey of Li *et al.* [1] catalogues a year of such breaks, nearly all exploiting plaintext-independent diffusion. For the confusion stage alone, Li and Lo [4] established the optimal number of known/chosen plaintexts needed to invert a permutation-only multimedia cipher, of order $\log_L(MN)$ for L intensity levels, the same logarithmic regime our attack achieves; the general quantitative theory of permutation-only cryptanalysis [12] bounds this effort, and recent chosen-plaintext breaks continue to recover equivalent keys from a handful of images [13], [14]. Alvarez and Li [5] set out basic requirements for chaos-based cryptosystems, including plaintext-dependent keystreams and sensitivity to the plaintext; LLEO violates these. The differential metrics we scrutinise were standardised by Wu *et al.* [15], who also note that NPCR/UACI quantify sensitivity to a one-pixel plaintext change, precisely the quantity LLEO fails to deliver despite its reported value.

The pattern is structural, not incidental: when both confusion and diffusion are fixed by the key the cipher is, as a function of the plaintext, an affine or monomial map, and composing more such stages changes only its coefficients [1], [4]. The decisive question is binary: does any key material depend on the image? Schemes that answer “no” (LLEO included) fall to an equivalent key, while those that bind the keystream to a per-image value resist, both sides of which our positive control demonstrates on the same construction (Section V).

What is new here. Prior equivalent-key cryptanalyses (Arroyo *et al.* [3], Li *et al.* [1], [2]) break one named scheme at a time by hand, each analysis bespoke to its target. Our contribution automates this process: a black-box decision pro-

cedure that, knowing nothing of a cipher’s internals, identifies the position-wise algebra (affine, XOR, or GF(2)-affine), recovers an equivalent key, and issues a held-out certificate we prove sound, and the few-query bound matches the Li–Lo permutation-only optimum [4]. Table I places our break beside prior equivalent-key cryptanalyses: like them it recovers an equivalent key from a handful of chosen plaintexts, and only this work additionally distils the single break into a reusable audit and a graded plaintext-dependence metric. The automated audit that runs this procedure across a corpus and real published schemes is developed in the extended version.

III. THE TARGET CIPHER

LLEO encrypts an $M \times N$ grayscale image X in three key-driven steps.

(1) *Substitution.* X is split into 16 horizontal strips. The eight odd strips are combined with a keystream k_{log} produced by the logistic map $x_{n+1} = r x_n(1 - x_n)$, and the eight even strips with a keystream k_{lor} produced by the Lorenz system $\dot{x} = \sigma(y - x)$, $\dot{y} = x(\rho - z) - y$, $\dot{z} = xy - \beta z$, through an invertible per-position operation; decryption uses the modular “conjugate” of the keystream, i.e. a multiplicative inverse.

(2) *Permutation 1.* The 16 horizontal strips are reordered by a Fisher–Yates shuffle (a standard method for randomly reordering a list).

(3) *Permutation 2.* The result is divided into 16 vertical strips and shuffled again, giving the ciphertext.

The four “decryption keys” are the two keystreams (d_1, d_2) and the two shuffle index lists (d_3, d_4) . The chaotic initial conditions are derived from the secret key and the shuffles are seeded from it; *no quantity depends on the image X* . Although the scheme is repeatedly called “asymmetric,” the same four keys both encrypt and decrypt: it is a symmetric cipher.

IV. CRYPTANALYSIS

A. Threat model

We work in the standard chosen-plaintext setting under Kerckhoffs’s principle (the attacker knows the algorithm; only the key is secret): the attacker can obtain the ciphertext of any image it chooses under a fixed unknown key, and seeks an *equivalent key*, a procedure that decrypts any ciphertext of the probed size, not the chaotic secret seed itself. Because the cipher is plaintext-independent, the same equivalent key also follows from enough known plaintext–ciphertext pairs [18]; we present the chosen-plaintext version as it is minimal.

B. The whole cipher is one fixed map

Index the image as a length- n vector, $n = MN$. Each LLEO step is linear over $\mathbb{Z}_{256}$ and fixed by the key: substitution multiplies coordinate p by a fixed unit $K[p]$ (odd, hence invertible mod 256), and the two block shuffles compose into

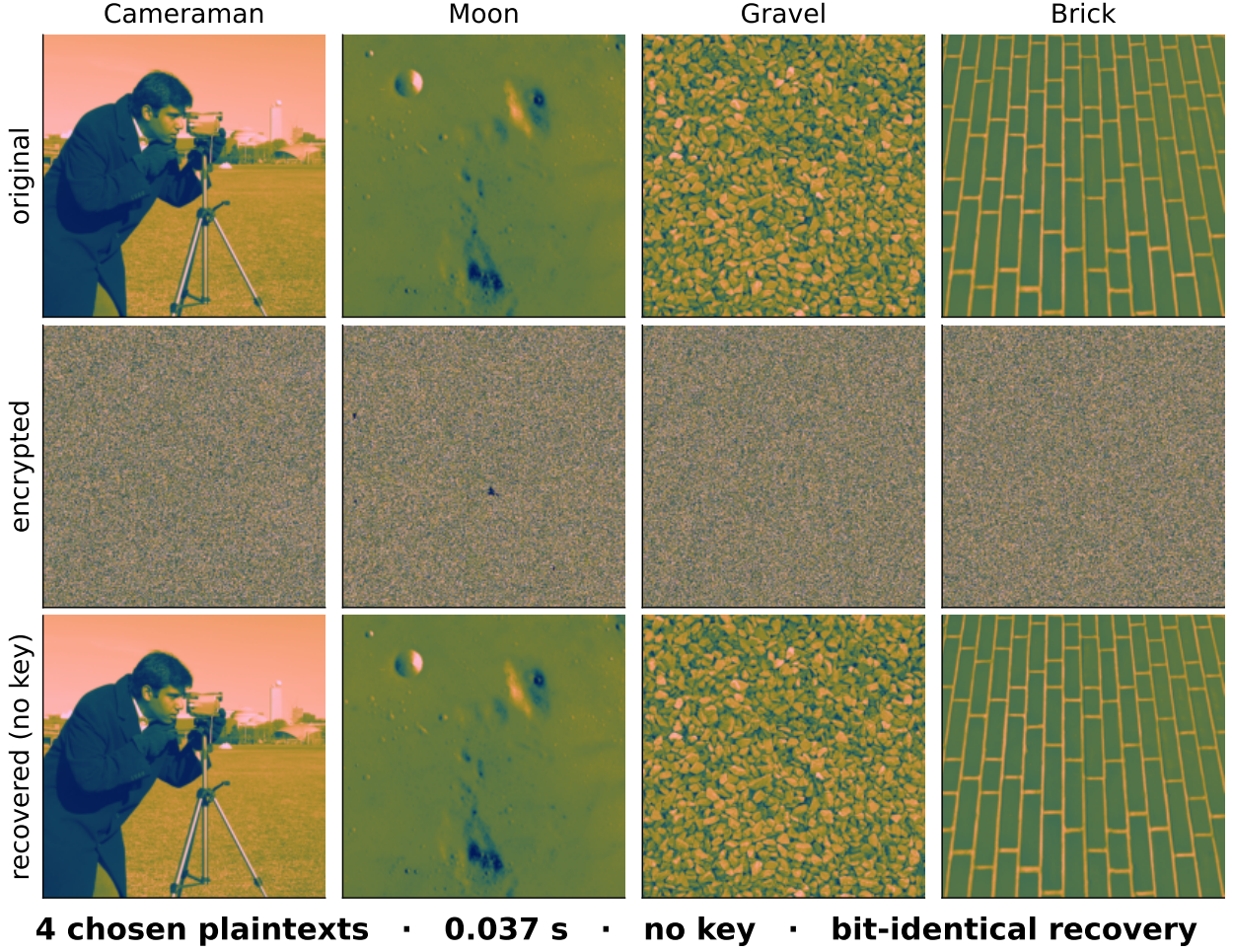


Fig. 2. The whole break in one view: four standard 512×512 images, their LEO ciphertexts, and our key-free recoveries: each bit-identical to the original, from 4 chosen plaintexts in 0.037s, without the key. Rendered in a perceptual colour map for legibility; the underlying images and every reported statistic are grayscale.

TABLE I

This break in the context of prior equivalent-key cryptanalyses of chaos / permutation image ciphers. “Chosen PT” is the number of chosen (or known) plaintexts; “equiv. key” means a key-free decryptor is recovered. Only this work distills the single break into a reusable audit and a graded plaintext-dependence metric (extended version).

Work	Target scheme	Chosen PT	Equiv. key
Li & Lo [4]	permutation-only	$\lceil \log_L MN \rceil$	perm.
Jolfaei <i>et al.</i> [16]	permutation-only	$\lceil \log_{256} MN \rceil$	perm.
Arroyo <i>et al.</i> [3]	total-shuffling	few	yes
Wen & Lin [17]	chaos+DNA (QCMDC)	4	yes
Recent [13], [14]	2020–2025 chaos	handful	yes
This work	LLEO [10]	4	full map

one position permutation P . Therefore

$$C[j] = K[s(j)] \cdot X[s(j)] \pmod{256}, \quad s = P^{-1}, \quad (1)$$

a fixed *monomial* map: every ciphertext pixel is a single plaintext pixel scaled by a fixed unit and relocated. Two consequences are immediate. There is no additive constant, since $E(\mathbf{0}) = \mathbf{0}$. And there is no diffusion: $C[j]$ depends on exactly one input coordinate $s(j)$, so a one-pixel plaintext change alters exactly one ciphertext pixel, which we exploit in Section V and visualize in Fig. 3.

Proposition 1. *If a permutation–diffusion cipher’s keystream and permutations depend only on the key, it realises a fixed affine map of the plaintext: the monomial (1) with each*

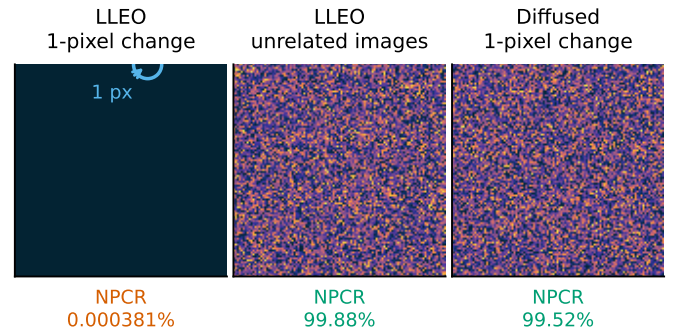


Fig. 3. The no-diffusion property, made visible (used quantitatively in Section V): the absolute ciphertext difference $|C_0 - C_1|$ (dark = unchanged, bright = changed), under one fixed key. **(left)** A one-pixel plaintext change alters *exactly one* ciphertext pixel (ringed), no diffusion, so the true one-pixel NPCR is $100/MN \approx 0.000381\%$, not the reported $\approx 99.6\%$. **(centre)** That $\approx 99.6\%$ is really the difference between two *unrelated* images (here 99.88%), which any near-uniform cipher attains. **(right)** Seeding the keystream from SHA-256 of the plaintext diffuses the *same* one-pixel change into full noise, what a one-pixel NPCR should measure.

$K[p]$ a unit in $\mathbb{Z}_{256}$ when the substitution is a modular multiply, or an affine map $C = Mx \oplus b$ over $\text{GF}(2)$ (with $b = E(\mathbf{0})$ possibly nonzero) when the substitution is XOR-based; composing any number of such rounds yields a map of the same form. In either case it is recovered from $1 + \lceil \log_{256} n \rceil$ chosen plaintexts: one all-ones (resp. all-zeros)

query fixes the substitution at every output position, and $\lceil \log_{256} n \rceil$ index-digit queries fix the permutation s .

Proof sketch. Each stage, multiplication by the key-derived K and the two block shuffles, is a fixed function of the key applied to pixel values or positions, so any composition is a fixed key-only map; over $\mathbb{Z}_{256}$ this is the monomial (1). The keystream is forced odd, so $K[p]$ is a unit and $x \mapsto K[p]x$ is a bijection with inverse $K[p]^{-1}$. Querying $X = 1$ gives $C[j] = K[s(j)]$; querying the t -th base-256 digit of the index gives $C[j] = K[s(j)] \text{digit}_t(s(j))$, and dividing by the recovered $K[s(j)]$ yields $\text{digit}_t(s(j))$; collecting all digits reconstructs s . If instead the substitution is XOR with a key-only keystream, the same argument over $\text{GF}(2)$ gives an affine map $C = Mx \oplus b$ with $b = E(\mathbf{0})$: the all-zeros query returns b and the index-digit queries return s , exactly as in a CBC-like XOR cipher (extended version). $\square$

C. Four images recover the full map

In plain terms: because nothing in the cipher depends on the picture, it always does the same two things: shuffle the pixel positions in one fixed way, and multiply each pixel by one fixed number. So feed it an all-ones image and it hands back the multipliers; feed it images whose pixels carry their own position numbers and it hands back the shuffle. About four such images, and the cipher is undone forever, without ever learning the key.

Formally, because K and s are fixed, two kinds of chosen plaintext suffice.

(a) *Multiplier, one query.* Encrypting the all-ones image 1 gives, by (1), $C[j] = K[s(j)] \cdot 1 = K[s(j)]$: the exact multiplier acting at every output position.

(b) *Permutation, $\lceil \log_{256} n \rceil$ queries.* Let P_t be the plaintext whose pixel i holds the t -th base-256 digit of its index i . Then $C[j] = K[s(j)] \cdot \text{digit}_t(s(j))$; multiplying by the already-known $K[s(j)]^{-1}$ yields $\text{digit}_t(s(j))$. Collecting all $\lceil \log_{256} n \rceil$ digits reconstructs $s(j)$ for every j , hence the full permutation.

The recovered pair (s, K) is an equivalent key; any ciphertext is inverted by $X[s(j)] = K[s(j)]^{-1} C[j]$. The total cost is

$$1 + \lceil \log_{256} n \rceil \text{ chosen plaintexts,} \quad (2)$$

i.e. 4 for a 512×512 image, independent of the nominal key length.

Concretely, a 512×512 image needs four queries, one all-ones image and three index-digit images, after which Algorithm 1 returns the full map in a single pass. Table II states the complexity and compares it to the optimal permutation-only bound of Li and Lo [4]: our query count is order-optimal and the cost is independent of the cipher's nominal key length, so the advertised keyspace is irrelevant; the equivalent key bypasses key recovery entirely [16].

Algorithm 1 Equivalent-key recovery for LLEO

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1:  $\hat{K} \leftarrow \text{Enc}(\mathbf{1}) \triangleright \hat{K}[j] = K[s(j)]$ : multiplier at output  $j$ 
2: for  $t \leftarrow 0$  to  $\lceil \log_{256} n \rceil - 1$  do
3:    $C \leftarrow \text{Enc}(P_t) \triangleright P_t[i] =$  the  $t$ -th base-256 digit of index  $i$ 
4:    $d_t[j] \leftarrow C[j] \cdot \hat{K}[j]^{-1} \bmod 256$  for all  $j$ 
5: end for
6:  $s[j] \leftarrow \sum_t d_t[j] 256^t \triangleright$  source index for each output  $j$ 
7: return  $\text{Dec}(C) : X[s(j)] \leftarrow \hat{K}[j]^{-1} C[j] \bmod 256$ 

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TABLE II

Attack complexity vs. the optimal permutation-only bound ($n = MN$ pixels, $L = 256$ levels).

	This attack	Optimal perm.-only [4]
Data (chosen plaintexts)	$1 + \lceil \log_{256} n \rceil$	$\lceil \log_L n \rceil$
Time	$O(n \lceil \log_{256} n \rceil)$	$O(n \lceil \log_L n \rceil)$
Memory	$O(n)$	$O(n)$

D. Why extra structure does not help

A composition of fixed key-only maps is again one, so extra rounds, chaotic maps, or block shuffles leave (1) intact (only K, s change) and the attack applies unchanged; it is moreover a *known*-plaintext attack whenever the available plaintexts span the index digits. The single property that would defeat it is a keystream that depends on X .

V. EXPERIMENTAL RESULTS

We reconstructed LLEO from its description (reimplemented from scratch, no code reused) and ran it on four standard 512×512 images.

A. Reproduction

Table III shows the reconstruction reproduces LLEO's claimed statistics: cipher entropy approaches the ideal 8 (Cameraman 7.9914, up from the plaintext's 7.2317) and adjacent-pixel correlation drops to near zero (-0.0073). Fig. 4 shows the flat cipher histogram and Fig. 5 the destroyed correlation. By the usual yardsticks the cipher looks strong.

B. The break

The attack of Section IV recovers the equivalent key in 4 chosen plaintexts and decrypts a fresh ciphertext pixel-exactly in 0.037 s at 512^2 (single core, Table IV; recovery is image-content-independent, so this time holds for any image), using no key. Fig. 2 shows the original, the LLEO ciphertext, and the key-free recovery, which is identical to the original.

C. Cost scales logarithmically

Because the attack needs only $1 + \lceil \log_{256} MN \rceil$ chosen plaintexts, its cost is nearly flat as the image grows (Table IV): 3 plaintexts at 128^2 , 4 at 512^2 , and still 4 at 1024^2 , each recovered exactly in well under a second. The key length the scheme advertises does not enter this cost.

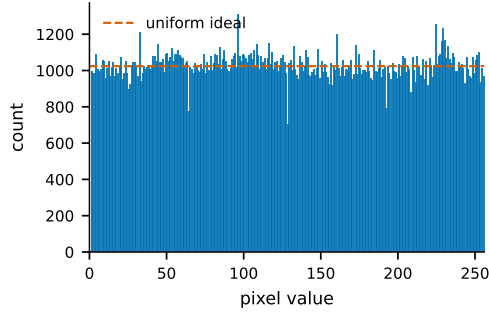


Fig. 4. Cipher histogram (Cameraman): near-uniform, yet the cipher is trivially broken.

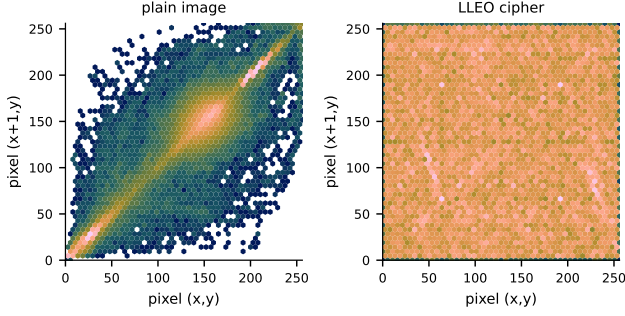


Fig. 5. Adjacent-pixel correlation as a hexbin density (every horizontal pixel pair, log counts): the plain image concentrates on the diagonal (neighbours are alike) while the LLEO cipher fills the plane uniformly (Cameraman). TABLE III

Reconstruction statistics and attack outcome (512×512). Every image is recovered pixel-exactly without the key.

Image	H plain	H cipher	corr.	CP	broken
Cameraman	7.2317	7.9914	-0.0073	4	yes
Moon	4.885	7.9316	0.0011	4	yes
Gravel	7.2531	7.9876	-0.0005	4	yes
Brick	5.4553	7.8966	-0.0007	4	yes

TABLE IV

Cost barely grows with image size: 4 chosen plaintexts still suffice at 1024×1024, each inverted in 0.254 s (random plaintext, single core).

Size	chosen PT	time (s)	recovered
128 × 128	3	0.003	exact
256 × 256	3	0.009	exact
512 × 512	4	0.037	exact
1024 × 1024	4	0.254	exact

D. The differential metric is illusory

LLEO reports $\text{NPCR} \approx 99.6\%$ as evidence of differential-attack resistance. But by (1) flipping one plaintext pixel changes exactly one ciphertext pixel (Fig. 3, left), so the true one-pixel NPCR is $100/MN = 0.000381\%$ (UACI 4e-06 %), five orders of magnitude below the reported value. The 99.6 % figure instead measures the difference between *unrelated* images, which for LLEO is 99.88 % (Fig. 3, centre); this is achieved by any cipher with a near-uniform output and says nothing about chosen-plaintext security. Table V contrasts each claimed property with what the cipher actually provides. Finally, the chaotic keystreams are themselves non-uniform (Fig. 6): a chi-square test (a standard check for whether values are evenly spread) rejects uniformity for the logistic keystream ($\chi^2 = 525839.6$, $p < 10^{-300}$, numerically zero).

The single break generalizes. Treated as a black-box structural audit, the same equivalent-key recovery decides

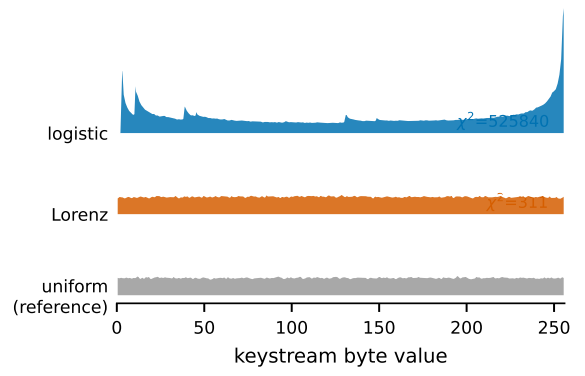


Fig. 6. Keystream byte distributions (ridgeline). The logistic keystream is sharply non-uniform (its invariant density spikes at the extremes) and even the Lorenz stream is measurably lumpy, whereas a secure stream cipher would match the flat uniform reference. A chi-square test rejects uniformity for both (χ^2 annotated).

TABLE V
Claimed vs. actual security of LLEO.

Property	Reported	Actual
Entropy (Cameraman)	≈ 7.997	7.9914 (genuine)
NPCR (1-pixel change)	$\approx 99.6\%$	0.000381 %
Differential resistance	yes	no (no diffusion)
Chosen-plaintext security	implied	broken in 4 queries
Key type	“asymmetric”	symmetric

plaintext-independence for an unknown cipher and certifies each verdict against held-out ciphertext; that audit, its application to real published 2023–2025 schemes, and a graded plaintext-keying-capacity metric separating fragile from genuine plaintext binding are developed in an extended version.

VI. DISCUSSION

LLEO illustrates a persistent failure mode: *good statistics are not security*. A flat histogram, near-ideal entropy, near-zero correlation, and a large inter-image NPCR are all satisfied by our fixed, key-only map, which we nonetheless break in 4 queries; testing plaintext sensitivity directly [5], [15] is what catches this. The scheme’s “asymmetric” label is also a misnomer: the same keys encrypt and decrypt, which is a labeling error rather than a structural one.

The remedies are standard. (i) Make the keystream depend on the plaintext, e.g. derive the chaotic initial conditions from a SHA-256 hash of the image, so a different plaintext yields an unrelated keystream. (ii) Add genuine diffusion, e.g. a chaining rule $C[i] = f(X[i], k[i], C[i-1])$, so one plaintext change propagates. With either, (1) is no longer fixed across images and the attack fails: it recovers 100.0 % of pixels against the key-only cipher but only 0.2518 % (chance) once the keystream is SHA-256-seeded from the image.

A. Limitations

One bound on our claims: we recover an *equivalent key* that decrypts any ciphertext of the probed size, not the underlying chaotic secret seed itself, the standard, defensible claim in this literature [1].

Disclosure. Consistent with responsible-disclosure practice, we emailed the authors of every scheme we

break (LLEO [10]) at submission, ahead of publication. We had received no response at the time of writing, and will incorporate any correction in the camera-ready. Our reproduction and attack code is available at [anonymized code repository URL; insert before submission].

VII. CONCLUSION

A 2025 chaos-based image cipher is completely broken by an equivalent-key chosen-plaintext attack in 4 chosen plaintexts and a fraction of a second, despite near-ideal entropy and a reported NPCR of $\approx 99.6\%$: good statistics are not security, and the cipher's differential metric, measured correctly, is 0.000381% . We recommend plaintext-dependent keystreams and real diffusion, and that statistical scores accompany, rather than replace, an explicit chosen-plaintext security argument. More broadly, the attack generalizes into a reusable structural audit that decides plaintext-independence, with a held-out certificate we prove sound; its application to real published schemes and a graded plaintext-keying-capacity metric are developed in an extended version.

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